

PRITISH SAHA

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Education

Indian Institute of Technology, Kharagpur

Kharagpur, West Bengal

B.Tech. in Manufacturing Science & Engineering + M.Tech. in Industrial & Systems Engineering

Expected completion: May'27

Relevant Courses: Safety Fundamentals of Generative AI | Operations Research | Probability & Statistics | Linear Algebra

Research & Engineering Experience

Research Fellow (MARS 4.0) | Cambridge AI Safety Hub | UK (Hybrid) | [\[Report\]](#)

Dec'25 – Present

With Prof. Fernando Rosas, studying Bayesian belief representations in transformers and extending the experiments to RL

- Built **exact Bayesian filters** to track hidden-state uncertainty from streams with complete or partial observations
- Decoded beliefs from **four-layer transformers** in fully observed pilots: probe $R^2 = 0.985\text{--}0.997$ after 100k updates
- Implemented solvers for partially observed tasks and changing rewards, with **115 tests** of Bayesian filtering and control

Data Science Intern | Axis Bank, Business Intelligence Unit | Mumbai | [\[Report\]](#)

May'26 – Jul'26

Built a PySpark pipeline to identify personal-loan fraud rings across 224.8M accounts and 1.31B transfer edges

- Pruned the working graph $\sim 20\times$ with time-respecting search, retaining connections for $\sim 80\%$ of applicants
- Delivered a four-signal fraud review queue, with **5.49 $\times$ mean lift** in the **top 1%** across three monthly cohorts

Research Intern | RAAPID INC | Remote | [\[Research\]](#) [\[Patent\]](#) [\[LoR\]](#)

Mar'25 – Jul'26

With Prof. Amitava Das, developed efficient LLM fine-tuning methods and quality checks for clinical entity extraction

- Prototyped **Triton kernels** for covariance estimates, matrix inversion and gradient updates in **LoRA fine-tuning**
- Integrated **CUDA streams and events** for asynchronous matrix updates, with **PyTorch fallbacks**
- Applied cleanup rules to **5,035 stored clinical predictions: 0.640 exact micro-F1** and **zero subword fragments**

Research Intern | Complex Networks Research Group, IIT Kharagpur | [\[Report\]](#) [\[LoR\]](#)

Jul'25 – May'26

With Prof. Pawan Goyal, studied how reward design affects mathematical reasoning during reinforcement learning

- Trained **Qwen2.5-Math-7B** with **GRPO** and **LoRA** on one H100, generating **16 solutions** per training problem
- Selected teacher-augmented pilot: **79.57%** MATH-500 mean correctness (**8 samples/problem**; **+4.70 pp** vs GRPO)
- A separate audit found shaped credit aligned with GRPO (cosine **0.979**), with no shaping signal on all-wrong groups

Publications & Preprints

Mixture of Chapters | Accepted, ICLR 2026 NFAM Workshop | [\[Code\]](#) | [\[Paper\]](#)

Mar'26

Tasmay Pankaj Tibrewal, **Pritish Saha**, Ankit Meda, Kunal Singh, Pradeep Moturi

- Co-developed sparse transformer memory; pretrained **150M–1B-parameter models** on up to **15B tokens**
- ARC-C accuracy fell **2.68 percentage points** after instruction tuning, versus **6.69** for a compute-matched baseline

GRIT: Geometry-aware PEFT | arXiv Preprint | [\[Code\]](#) | [\[Paper\]](#)

Jan'26

Pritish Saha, C. Rajbangshi, R. Goyal, M. Goyal, A. Deo, B. Roy, N. D. Singh, R. Goswami, A. Das

- Developed a **LoRA** method that uses loss curvature to guide updates and adapt its effective rank
- Reached **66.19% on GSM8K** vs LoRA's 63.38% with Llama-3.1-8B and **27.8% fewer** effective adapter parameters

Competitions

3rd Place | General Championship Data Analytics, IIT Kharagpur | [\[Report\]](#)

Mar'26

Captained Patel Hall's team in building a conversational analytics dashboard for database queries and KPI discovery

- Built **text-to-SQL** with retrieval and repair: **90% execution accuracy** on a **50-query** BIRD mini-dev sample
- Automated KPI discovery with a **LangGraph pipeline**; generated **874K synthetic rows** across **58 client profiles**

4th Place | ISRO GeoNLI, Inter IIT Tech Meet 14.0 | [\[Report\]](#)

Oct'25 – Dec'25

Co-designed a remote-sensing assistant for image captioning, object grounding and visual question answering

- Designed a **Qwen3-VL-8B** curriculum using **LoRA SFT** and **DPO** for satellite captioning and question answering
- Paired **SAM3** with image tiling and VLM refinement to count objects, estimate areas and answer spatial queries

National Finalist | American Express Campus Challenge, Decision Science | [\[Report\]](#)

Jul'25

Built a personalized offer-ranking system from customer histories, combining tree models with a transformer

- Reached **0.59 MAP@7** on final evaluation; trained a transformer to correct errors from **LightGBM** and **XGBoost**
- Generated **3K+ history features** with point-in-time joins and customer-grouped validation to limit leakage

Technical Skills

Languages & Systems: Python, C++, SQL, CUDA, Triton, PySpark, Linux, Docker, SLURM, Git

ML & Post-training: PyTorch, JAX, Transformers, LoRA/QLoRA, TRL, vLLM, FSDP/DTensor, FlashAttention-2

Extra-Curricular Activities

Karate black belt and tournament participant | Patel Hall interhall football and water polo teams | NSS village volunteer